Adi Karthikeya Sunkara Balasubramanyam

Data Scientist | **AI/ML Engineer** | **Data Engineer** | **Data Analyst**

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Professional Summary

M.S. Data Science graduate with hands-on work in machine learning, LLM and RAG systems, agentic AI, and SQL analytics. Databricks and Anthropic certified, daily AI-assisted development with Claude Code, 4th of 200 at an agentic AI hackathon.

Experience

Data Scientist Intern Sep 2026 - Present

Worldlink *Frisco, TX*

- Collaborate with senior data scientists on data preparation, model testing, and evaluation across AI/ML projects in Python.
- Build automation scripts and monitoring dashboards, and manage code, design notes, and documentation in team repositories.
- Present progress in daily stand-ups and weekly leadership reviews, using Jira and Office 365 for sprint planning.

Graduate Teaching Assistant Jan 2025 - May 2026

University of Houston *Houston, TX*

- Supported 120+ students in Pattern Recognition and Machine Learning and Engineering Data Analytics courses.
- Led debugging sessions on Python, SQL, and scikit-learn workflows: data cleaning, EDA, feature selection, model evaluation.
- Reworked assignment rubrics for CEA 4341 to speed up grading turnaround across the cohort.

Data Science Intern, NSF May 2025 - Jun 2025

University of Houston *Houston, TX*

- Selected for a nationally funded 5-week NSF program; profiled and transformed multi-source geophysics and energy datasets.
- Built facies classifiers (KNN, Random Forest, Decision Tree) on well-log data, scoring 0.8024 accuracy in a national competition.
- Applied PCA and K-Means to Tellus geophysics data; trained a TensorFlow rock-fertility model on zircon geochemistry.
- Built Tableau dashboards and QGIS maps from EIA and U.S. Census data for geothermal and solar potential reporting.

Certifications

Databricks - Certified Generative AI Engineer Associate, Oct 2026 | **Anthropic** - Claude Certified Architect (Foundations), Sep 2026

Skills

Languages: Python, SQL (PostgreSQL, MySQL, BigQuery), R, Java (undergrad), Data Structures and Algorithms (coursework)
ML & AI: PyTorch, TensorFlow, HuggingFace, scikit-learn, XGBoost, LightGBM, SHAP, MLflow, LangChain, LangGraph, Google ADK, RAG, ChromaDB, Vector Search, Prompt Engineering, Gemini, OpenAI APIs, BERT/RoBERTa/DeBERTa, Fine-Tuning
Cloud & MLOps: GCP (Cloud Run, BigQuery, Vertex AI), AWS (S3, EC2), Databricks (Vector Search, Model Serving, Unity Catalog), Docker, CI/CD, Git, Linux, AI-assisted development (Claude, Claude Code)
Backend & Visualization: FastAPI, Flask, REST APIs, Streamlit, Power BI, Tableau, pandas, NumPy, Django
Statistics: Regression, Classification, Hypothesis Testing, PCA, Cross-Validation, Time-Series, A/B Testing

Projects

RareDx, Agentic Rare Disease Diagnostic Assistant | *LangGraph, Neo4j, ChromaDB, Groq* Mar 2026 - May 2026

- Group project: built an Orphanet ingestion pipeline into ChromaDB with BioLORD, lifting Top-1 retrieval from 10% to 92.5%.
- Calibrated 7 retrieval strategies (280 evaluations, Brier 0.068, ECE 0.051), reaching Recall@5 100% with zero hallucination.
- Diagnosed the retrieval failure, built the ablation framework, and curated the 40-case adversarial benchmark.

ShopperAI, Agentic AI Shopping Assistant | *Google ADK, Gemini, GCP, FastAPI, Docker* Dec 2025

- Built a deterministic, stage-based agentic shopping assistant guiding users through product discovery and virtual try-on.
- Deployed as containerized FastAPI/Docker services on GCP Cloud Run; placed 4th globally of 200 teams at the ODSC AI and Google Cloud Agentic AI Hackathon.

Predictive Match ML Dashboard | *XGBoost, LightGBM, SHAP, Streamlit, SQL/Pandas* Jun 2026

- Engineered a 52-feature pipeline over 60,000 profiles, trained stacked ensembles, and shipped a Streamlit SHAP dashboard.
- Built the dataset from 59,944 real OKCupid profiles with SQL and pandas, cleaning and encoding features for the ensemble.

Question Answering System (NLP) | *BERT, RoBERTa, DeBERTa, Flan-T5, PyTorch, SQuAD v2* Sep 2025 - Dec 2025

- Fine-tuned four transformer architectures on SQuAD v2, improving Exact Match from 34.9% to 75.3% and F1 to 78.4%.
- Benchmarked BERT, RoBERTa, DeBERTa, and Flan-T5 in PyTorch, comparing extractive and generative answering.

Olist E-Commerce Analytics | *PostgreSQL, SQL, Tableau* May 2026

- Modeled 100K orders in PostgreSQL with 9 SQL queries and a Tableau dashboard; late delivery drops reviews 4.3 to 1.7 stars.
- Answered 5 business questions on revenue concentration, delivery performance, and repeat purchase rate across a 7-table schema.

Education

University of Houston Houston, TX

M.S. Data Science, GPA 3.9/4.0 *May 2026*

Amrita Vishwa Vidyapeetham Bengaluru, India

B.Tech, Electronics and Communication Engineering, GPA 3.50/4.00 *May 2024*